

Credit Card Customer Segmentation

Who are our customers —
and what do they actually need?

8,950 customers • 17 behavioral features • 4 customer segments

Part 1 — Getting to Know the Data

The dataset

8,950 credit card customers, each described by 17 numeric features — balances, purchases, cash advances, payments, credit limits, and account tenure. All numeric, no categories to convert, ready for clustering.

What	Details
Total customers	8,950
Features per customer	17 (balance, purchases, payments, cash advances, etc.)
Missing values	CREDIT_LIMIT: 1 row dropped MINIMUM_PAYMENTS: 313 rows filled with 0

Handling missing values

The 313 missing minimum payment values were not a data error — they came from low-activity customers who simply had no payment due. Their average balance was \$555 vs \$1,601 for everyone else. Filling with 0 reflects that truth.

Fixing the skewness

Credit card data is heavily right-skewed — a few big spenders distort the entire scale. We applied a log transformation to the 13 most skewed features so the algorithm focuses on patterns, not outliers.

Feature	Before	After
MINIMUM_PAYMENTS	13.81	-1.29
ONEOFF_PURCHASES	10.04	0.19
PURCHASES	8.14	-0.76
CASH_ADVANCE	5.17	0.26
BALANCE	2.39	-0.86

Key finding before clustering: Purchases and Cash Advances have a correlation of -0.50. Customers who buy things rarely take cash advances — and customers who take cash advances rarely buy things. Two fundamentally different behaviors already visible in the raw data.

Part 2 — Finding the Right Number of Groups

Reducing dimensions with PCA

Many of our 17 features were correlated with each other — like cash advance amount and cash advance frequency, which say nearly the same thing. PCA compresses them into 7 independent summary features that still carry 89.5% of all the information.

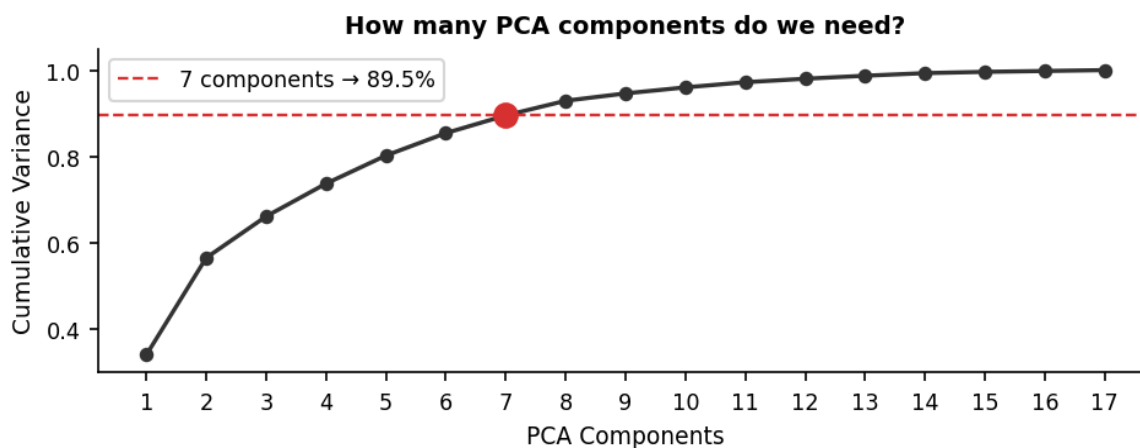


Figure 1 — 7 PCA components retain 89.5% of variance

How many groups?

We tested $k=2$ through $k=10$. The Elbow Method and Silhouette Score both pointed to $k=4$ as the sweet spot — meaningful separation without over-splitting.

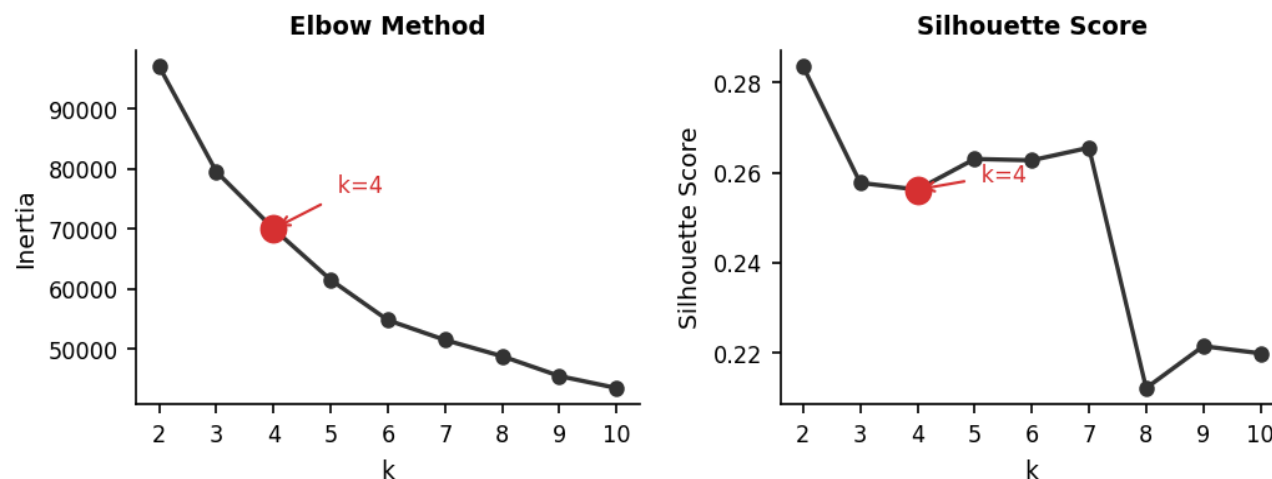


Figure 2 — Elbow and Silhouette both support $k=4$

We also compared $k=4$ vs $k=6$. The silhouette difference was only 0.007 — negligible. But $k=6$ produced one tiny group of 785 customers and three near-identical mid-sized groups. Four groups gave cleaner, more actionable segments.

Part 3 — The 4 Customer Types

Four distinct groups emerged from the data. Each one has a clear behavioral signature and a different business implication.

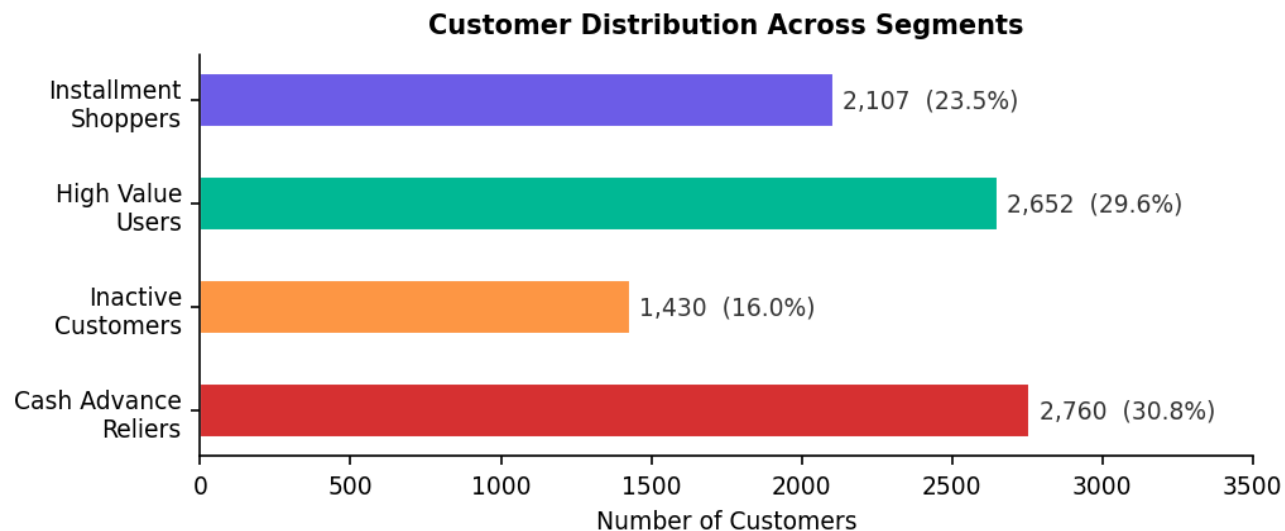


Figure 3 — Segment sizes

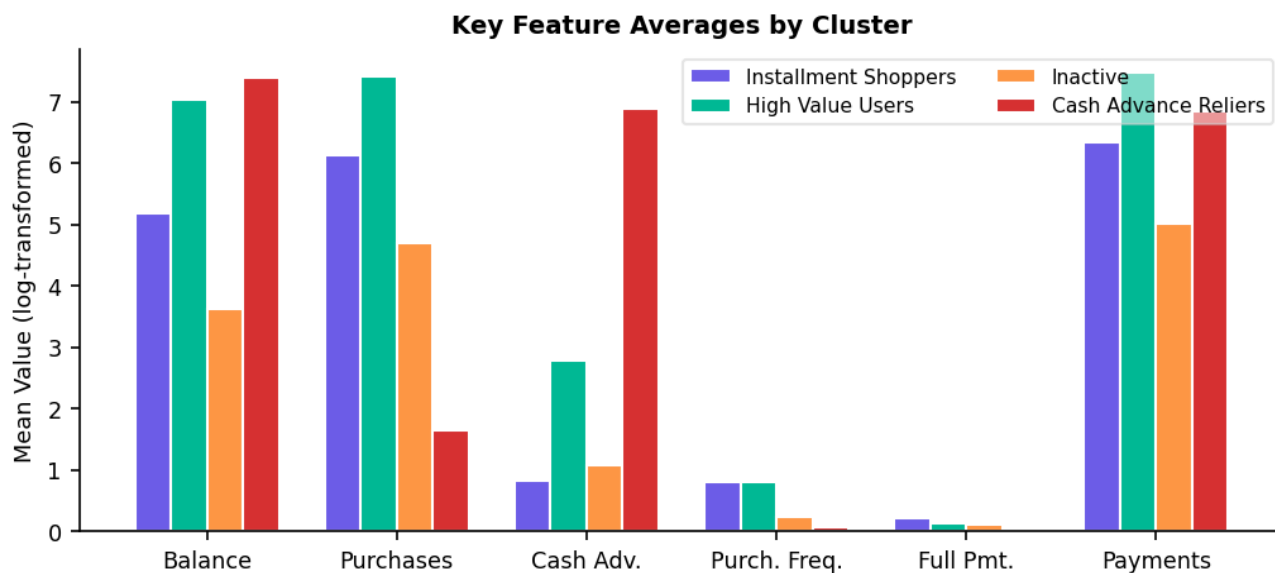


Figure 4 — Mean feature values by cluster (log-transformed)

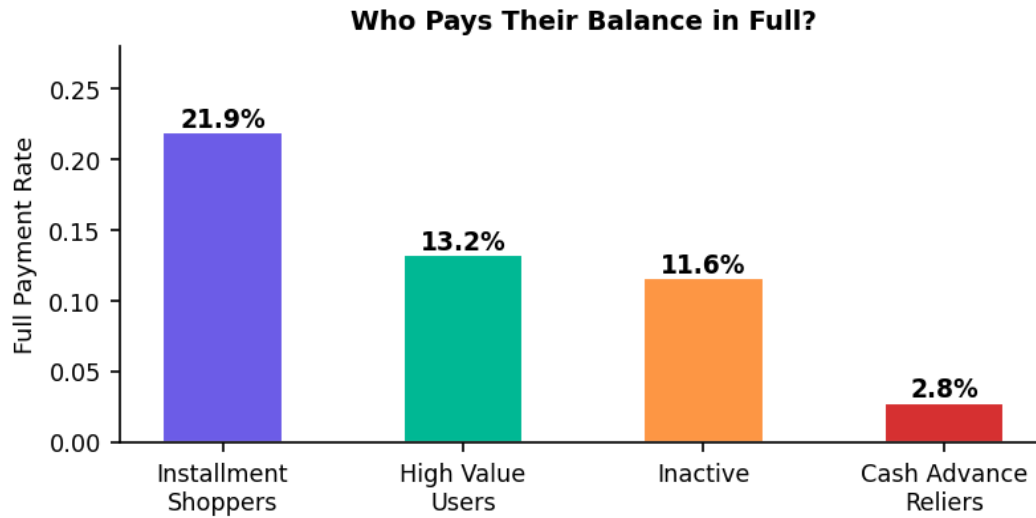


Figure 5 — Full balance payment rate by cluster

The gap between Cluster 0 (21.9% full payment) and Cluster 3 (2.8%) tells the whole story. Installment Shoppers are financially disciplined. Cash Advance Reliers are carrying growing debt. This single metric captures the risk difference between the two ends of the customer spectrum.

Cluster Profiles in Detail

Cluster 0 Installment Shoppers

2,107 customers • 23.5%

"I prefer to spread my payments — I'm careful with money."

Reliable, consistent, and disciplined. These customers shop regularly but almost exclusively through installment plans. They have the highest full-payment rate of any group and almost never touch cash advances.

- High installment purchases (6.03) — one-off purchases nearly zero (1.20)
- Shop frequently — purchase frequency of 0.80
- Highest full payment rate: 21.9%
- Minimal cash advance use — score of just 0.82

What to do: Reward loyalty with installment-specific promotions and 0% APR deals. Cross-sell savings products — they are financially healthy enough. These are your low-risk, long-term customers. Protect them.

Cluster 1 High Value Users

2,652 customers • 29.6%

"I use my card for everything — and I spend big."

The most valuable segment. Heavy spenders across both one-off and installment purchases, highest credit limits, and largest payments. Fully engaged with their card in every way.

- Highest total purchases (7.41) and one-off purchases (6.77)
- Highest credit limit (8.52) and payments (7.48)
- Very frequent shoppers — purchase frequency of 0.81

What to do: Treat them like VIPs. Premium card upgrades, travel rewards, exclusive offers. Losing one of these customers hurts more than losing five from other segments.

Cluster 2 Inactive Customers

1,430 customers • 16.0%

"I have the card, but I barely use it."

Present but disengaged. They rarely shop, carry low balances, and show little activity across almost all features. Not in trouble — just not connected to the product. A missed opportunity.

- Low purchase frequency — just 0.25
- Balance frequency of only 0.52 — account often sits empty
- Not heavy cash advance users either — broadly disengaged

What to do: Run re-engagement campaigns with time-limited spending bonuses. Send personalized offers based on past purchases. If still no engagement after outreach, flag as churn risk and manage the account cost.

Cluster 3 Cash Advance Reliers

2,760 customers • 30.8%

"I use my card to get cash when I need it — I'm stretched thin."

The largest group and the most concerning. These customers use their card almost exclusively for cash advances, carry the highest balances, and almost never pay in full. This pattern is consistent with financial stress.

- Cash advance score of 6.89 — far above every other group
- Almost no purchases — score of just 1.65
- Lowest full payment rate: only 2.8%
- Highest revolving balance (7.39)

What to do: Offer debt consolidation loans and flexible repayment plans. Set early-warning triggers for escalating cash advance frequency. Review credit limits for the highest-risk users. The goal is to help them — which also protects the bank.

Part 4 — How We Validated the Results

Multiple visualization methods confirmed the 4 natural groups

After clustering, we visualized the data using UMAP and t-SNE — two methods that find structure in data without being told what to look for. Both independently revealed 4–6 clearly separated islands, confirming the groups are real.

Method	What it showed
PCA 2D	Cluster 3 (cash advance) was the most cleanly separated. Some overlap between groups 0 and 2 is expected when compressing 7 dimensions to 2.
UMAP	4–6 distinct blobs with clear space between them — groups exist naturally in the data without any labels.
t-SNE	Visible cluster separation with well-defined neighborhoods — further confirmed K-Means boundaries were meaningful.

We also tested GMM — K-Means won

Gaussian Mixture Models allows more flexible cluster shapes. We ran it with k=4 and compared directly to K-Means.

Algorithm	Group sizes	Verdict
K-Means	2,107 / 2,652 / 1,430 / 2,760	Balanced and interpretable — chosen as final model
GMM	1,691 / 1,695 / 1,829 / 3,734	One group absorbed 42% of customers — too dominant

GMM merged too many customers into one group of 3,734. The other three groups were suspiciously similar in size — a sign of one natural cluster split artificially. K-Means gave more balanced and more actionable results.

Part 5 — What This Means for the Business

Segment snapshot

Segment	Size	%	Revenue Potential	Risk	Priority
Installment Shoppers	2,107	23.5%	Medium–High	Low	Retain & reward
High Value Users	2,652	29.6%	Very High	Low	Premium programs
Inactive Customers	1,430	16.0%	Low (now)	Medium	Re-engagement
Cash Advance Reliers	2,760	30.8%	Low	High	Risk management

Three things this analysis is really saying

Nearly a third are financially stretched

30.8% of customers rely on cash advances with near-zero full repayment. This is not a niche edge case — it is the single largest segment and a systemic risk.

The top 30% carry the revenue

High Value Users generate the most purchases and payments. Protecting and growing this segment should be a top strategic priority.

16% are an untapped opportunity

Inactive customers are not lost — they just need the right offer at the right time. Re-engagement has low cost and meaningful upside.

Recommendations by segment

Cluster 0 — Installment Shoppers

Installment loyalty rewards • 0% APR promotions • Cross-sell savings products • Proactive retention outreach

Cluster 1 — High Value Users

Premium card upgrades • Tiered cashback programs • Luxury & travel merchant partnerships • Dedicated relationship management

Cluster 2 — Inactive Customers

Time-limited re-engagement bonuses • Category-specific personalized offers • Fee waivers tied to minimum spend • Churn risk flagging

Cluster 3 — Cash Advance Reliers

Debt consolidation loan offers • Early-warning outreach triggers • Financial wellness tools • Credit limit reviews for highest-risk users

Final Thoughts

What we found, in plain language

We started with a spreadsheet of 8,950 customers and 17 numbers per customer. No labels. No instructions. Just raw behavioral data.

By the end, we had four real human stories: the careful installment shopper, the high-flying big spender, the quietly disengaged, and the cash-stretched reliever. None of these groups needed to be invented — they were already there in the data, waiting to be found.

The machine learning confirmed what good customer intuition would probably guess: people use credit cards for very different reasons, and treating them all the same is a missed opportunity — both commercially and in terms of helping customers who may be struggling.

The most important number in this analysis is 30.8% — the share of customers relying on cash advances with almost no full repayment. That is not a data point to scroll past. It is a call to action.

Where to go from here

- Build a live scoring tool that assigns new customers to a segment at sign-up
- A/B test marketing strategies per segment and measure real uplift in spend and retention
- Track customers who shift between segments over time — moving toward Cluster 3 is an early warning sign
- Build an interactive dashboard so business teams can explore segments without running code